Mobile Application Development

Types of Mobile Application

Lecture # 1

Concepts you should be good at for this course

- Object Oriented Programming
  - Polymorphism
  - Abstraction
  - Inheritance
  - Encapsulation
- Advance Computer Programming
- Java

Todays Agenda

- Mobile Apps
  - Native Apps
  - Mobile Web Apps
  - Hybrid Apps
  - Cross Mobile Apps
- How android works
  - Architecture

Native Apps

"Native mobile apps are exclusively built for a specific type of Operating system. They are called native because they are native to a particular device or platform", e.g.

- Whatsapp android app
- Gmail android app
- Google maps android app
- Apple maps IOS app
- Safari browser IOS app
- iCamera IOS app

The native apps are design and developed usually on dedicated platform, e.g.

- Xcode and Objective C for IOS
- Eclipse/Android studio and Java for Android
- C# for Windows

Native apps can be accessed via respective app stores, i.e., Android apps on Google Play Stores, iOS apps on App Store

Native Apps

- **Advantages of native apps:**
  - Natives are very fast.
  - Easily distributed in google apple app stores.
  - Easily interact with any feature of the phone.

- **Disadvantages of native apps:**
  - Built for a single platform
  - Languages like swift and java used to build these types of apps are hard to learn.
  - Hard to maintain.

Mobile Web Apps

- These are the web applications to deliver web pages on web browsers running on mobile devices.
- These are web-based mobile apps that do not get installed on your handheld mobile device and are run on web-hosted servers.
- Mobile web apps typically use HTML, CSS, Javascript, JQuery web technologies.
- They cannot access all features of native device functionality(camera, calendar, geolocation, etc.)
- Example:
  - Facebook web-app
  - Instagram web-app

Mobile Web Apps

- **Advantages of web apps:**
  - Reduced business cost.
  - No installation needed.
  - Better reach as it can be accessed from anywhere.
  - Always up-to-date.

- **Disadvantages of web apps:**
  - Dependent on internet speed.
  - Interface not that sophisticated.
  - Security risk.

Hybrid Apps

- A (hybrid app) is a software application that combines elements of both native apps and web applications.
- Hybrid apps are essentially web apps that have been put in a native app shell.
- Once they are downloaded from an app store and installed locally, the shell is able to connect to whatever capabilities the mobile platform provides through a browser that's embedded in the app.
- The browser and its plug-ins run on the back end and are invisible to the end user.
- Examples:
  - Facebook Android App
  - Instagram Android App

Hybrid Apps

- **Advantages of hybrid apps:**
  - Easy to build
  - No browser needed
  - Can usually access device utilities using an API

- **Disadvantages of Hybrid apps:**
  - Slower than native apps
  - More expensive than web apps
  - Less interactive than native apps

Cross Mobile App

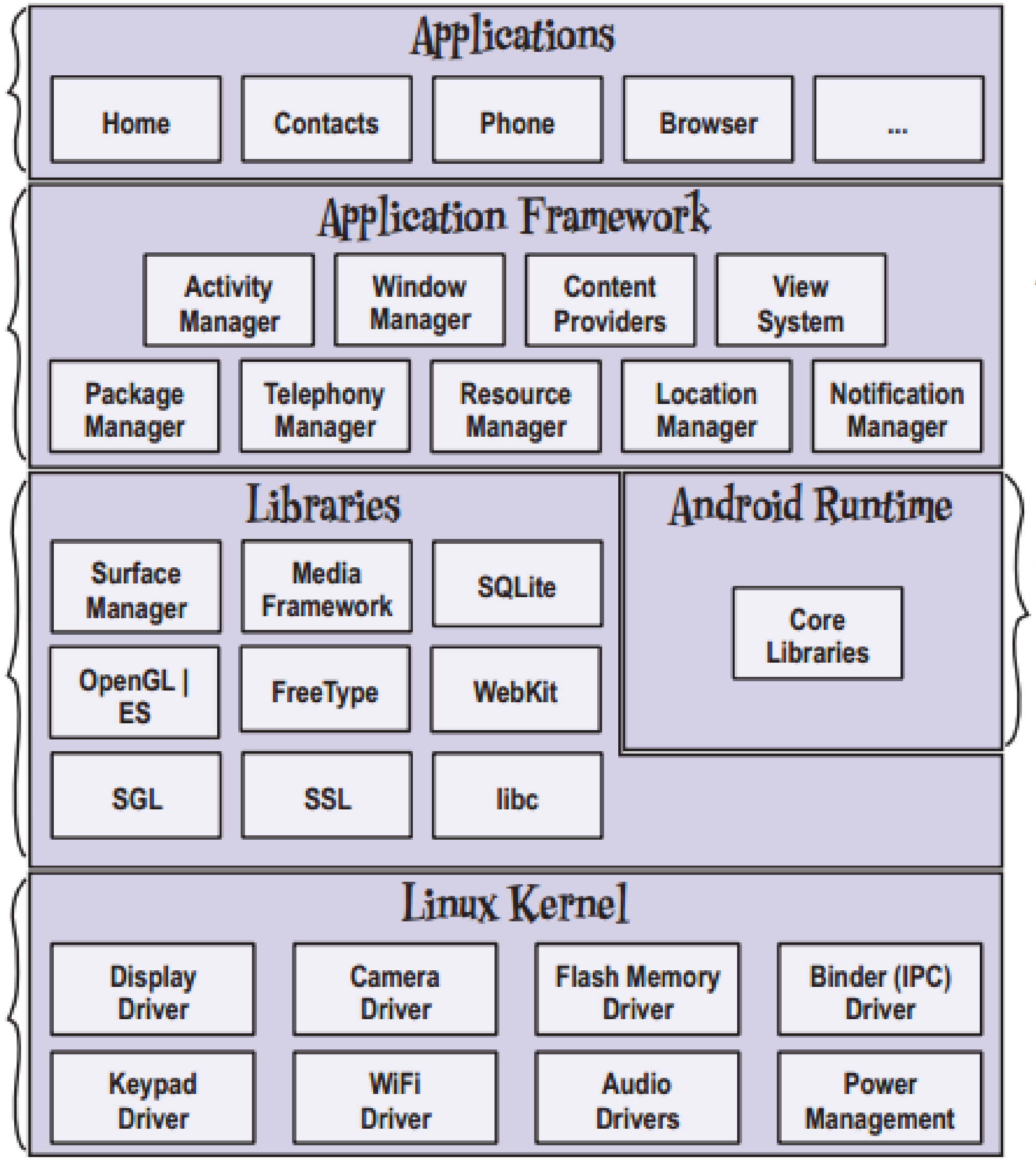
- Cross-platform mobile apps are developed using an intermediate language, such as Javascript, that is not native to the device’s operating system.
- This means that some, or all, of this code can be shared across target platforms, for instance, across both iOS and Android.
- Cross-platform apps are different to HTML5 hybrid apps as hybrid apps usually incorporate a mix of native app and mobile app concepts.
- Cross-platform apps are developed with Xamarin, Appcelerator, React Native, NativeScript, and Flutter.

Cross Mobile App

- **Advantages**
  - Code can be shared between different versions of the apps across devices (possibly up to 80%)
  - The User Interface is rendered using ‘native’ controls, so UI performance can be as fast as native.
- **Disadvantages**
  - Not all code can be shared, so some native code may need to be written
  - Access to the device, operating system features and interface customization rely on the framework or plugin support

Android Architecture

- Android uses Linux for its device drivers, memory management, process management, and networking.
- However you will never be programming to this layer directly.
- The next level up contains the Android native libraries.
- They are all written in C/C++ internally, but you'll be calling them through Java interfaces.



Android Architecture

- Next is the Android runtime, including the Dalvik Virtual Machine.
- Dalvik runs dex files, which are converted at compile time from standard class and jar files.
- The next level up is the Application Framework layer.
- Parts of this toolkit are provided by Google, and the other parts are extensions or services that you write.
- The most important component of the framework is the Activity Manager, which manages the life cycle of applications and a common "back-stack" for user navigation.
- Finally, the top layer is the Applications layer.
- Most of your application will live here, along side built-in applications such as the Phone and Web Browser.

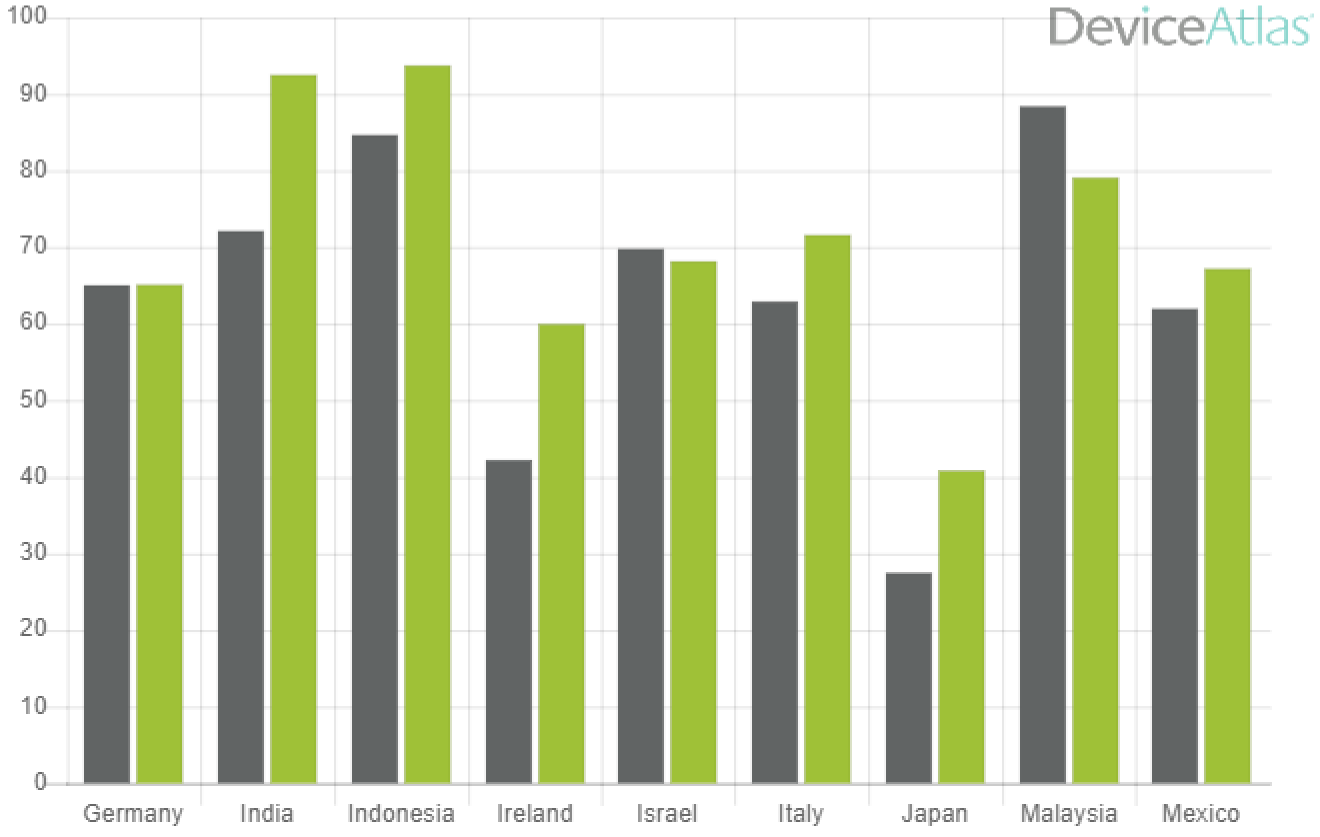
Challenges

- Limited resources
  - RAM
  - Memory
  - Battery
- Limited screen space
  - Small screen
  - Variation in screens
  - Webpages design
  - Hard to select small objects
- Different usage pattern

Why Android Platform?

- Fastest growing smartphone platform

- Google Android and Apple iOS have **98% of the global market share** for operating systems.
- Android's market share **will reach 87% in 2022**, forecasts suggest.



Android Development Tools

To set up the development environment, we need:

1. Java Development Kit (JDK 1.5+, 1.6 is preferable)
2. Eclipse IDE
3. Android Studio

To run android App

1. Android Emulator
2. Android Device